

Amey Ambade

Senior Data Scientist

Generative AI & MLOps | 7+ Years Production ML | \$20M+ Client Impact | SE Radio Podcast Host

ameyambade@gmail.com | (646) 961-8378 | [linkedin/ameyambade](https://www.linkedin.com/in/ameyambade) | [github/ameyxd](https://github.com/ameyxd) | heyamey.com

PROFESSIONAL SUMMARY

Senior AI / Lead ML Scientist with 7+ years delivering end-to-end, production ML software systems and client-facing AI consulting engagements that generated \$20M+ in measurable client value across energy, pharma, insurance, retail, and fintech. Deep expertise across Generative AI (LLMs, agentic systems, RAG, SLMs), MLOps, and software development, with patents in applied AI/ML. Proven leader of cross-functional teams and recognized contributor to the AI standards community (IEEE Senior Member, SciPy Proceedings Chair), translating research into deployed products and complex ML into executive-level business outcomes.

EDUCATION

M.S. Computer Science (Machine Learning Specialization)

New York, NY | 2019

COLUMBIA UNIVERSITY

B.Tech. Computer Science and Engineering

Indore, India | 2017

INDIAN INSTITUTE OF TECHNOLOGY (IIT)

WORK EXPERIENCE

SENIOR DATA SCIENTIST | DATAIKU

Houston, TX | 2026 – present

- Lead client-facing AI consulting engagements across **oil & gas, pharma, retail, insurance, fintech, and services**, currently managing **five concurrent client accounts**, scoping and co-developing production-grade **GenAI and agentic solutions** on the Dataiku platform, spanning hands-on solution delivery, pre-sales scoping, user enablement, and technical demos.
- Delivered a production AI deployment for a **Fortune 500 oil & gas client**, architecting a Dataiku-based solution used by **70 business intelligence users**, extending prior edge ML work into a centralized analytics and **LLM Mesh**-driven decisioning environment.
- Delivered **four production use cases** for a **Fortune 500 fintech client**: **merchant sales forecasting**, **financial entity mapping**, a **Financial Analyst Assistant LLM agent** on Dataiku's LLM Mesh, and a **Next Best Offer recommendation engine** combining the ALS-based recommender plugin with a complementary ranking algorithm, spanning classical ML, entity resolution, agentic GenAI, and recommender systems within a single account.
- Delivered a **multimodal RAG agent** for a global pharma client supporting **patent analysis and document retrieval**, with active development on downstream agents for **LLM-based clinical and doctor review tagging**. Tech: **LangChain, LangGraph, Cohere embeddings, vector databases**.
- Delivered a Dataiku-based POC for a **large insurance client**, projected to serve **200+ underwriting users** with an estimated **80% reduction in risk modeling cycle time**; shipped a custom plugin integrating **Moody's Risk Modeler APIs** into client workflows.
- Authored and shipped **custom Dataiku plugins** now part of the **proprietary plugin ecosystem**, including a content-based and collaborative filtering recommender (**ALS**) and client-specific API integrations, used in client production environments.
- Authoring an internal asset on **Cohere-based multimodal embeddings for semantic search**, to be published to Dataiku's internal best-practice library.
- Delivered **two technical workshops** on Dataiku's GenAI and agentic capabilities to **250+ external client end-users**; coach junior data scientists on solution design and client engagement, and conduct technical interviews for incoming junior data scientists.

LEAD DATA SCIENTIST - ML | EVOLEAP LLC

Houston, TX | 2025

- Implemented Model Context Protocol (MCP) Servers for AI-based API development within proprietary SaaS offerings like Flotools, Evoke, Obot, etc.
- Led development of sensor fusion transmission group filters using machine learning algorithms for onsite deployment on real-time data.

SENIOR DATA SCIENTIST - MACHINE LEARNING | SLB (SCHLUMBERGER)

Houston, TX | 2023 – 2025

- Led cross-functional client teams delivering **LLM** based AI solutions for Fortune 500 energy clients, managing full project lifecycle from requirements gathering to production deployment, resulting in 15% operational efficiency gains across 50+ client sites, including deployment of finetuned SLMs for **IIoT Edge** applications.
- Implemented CompetitorIQ, an LLM-based document analysis and agentic framework using OpenAI APIs and **RAG** architectures, reducing manual competitive research time by **40%** for business development teams and enabling real-time insights from unstructured data about company competitors.
- Generated **\$3M (Q1 2024) SaaS revenue** delivering research-derived LLM applications integrated with internal DS orchestration platforms and external data lakes. Tech: **Python, Dataiku, Azure Foundry**.
- Built and productized the **SLB Vision Analytics suite** (YOLOv5, transfer learning, CNN) for live video: fire/smoke detection, line of fire, people counters, PPE detection, pose tracking — **\$4.5M revenue, 45% field safety improvement, 35% PPE risk reduction, and 30% surveillance time reduction**.
- Led a 3-person team to deliver an **end-to-end ML service** for dynacard **image classification** and real-time control optimization across **100+ wells**, resulting in **\$10M+ savings, 60% faster detection, and 19% downtime reduction**. Tech: **TensorFlow, Docker, Flask, IoT Edge**.
- Designed and deployed **Smart Surveillance** for hydrate detection (**\$2M+ revenue, +15% production, -12% labor**).

DATA SCIENTIST | SCHLUMBERGER

Houston, TX | 2018 - 2023

- Developed **failure and anomaly prediction** models for Prognostic Health Management, implemented ML backend infrastructure and drove collaboration by partnering with field, client, and product teams to build production rate visualization with machine learning. Tech: **Azure ML, TensorFlow, Dataiku**
- Built a **deep learning**-based CNN model to classify erosion levels in mechanical tools and deployed a scalable cloud-based API, increasing precision by 8% and reducing detection time by 10%. Tech: **TensorFlow, Docker, Flask**
- Built a health analyzer application using **unsupervised learning** to predict tool failures to aid scientists and engineers. Tech: **Python, TensorFlow, FastAPI**

GRADUATE RESEARCHER | COLUMBIA UNIVERSITY, COLUMBIA LAW SCHOOL

New York, NY | 2018

- Built an attention-based encoder-decoder recommender (YouChoose CNARM), and co-developed NLP models for a contractual damage prediction engine.

VOLUNTEER EXPERIENCE

PROCEEDINGS CHAIR | SciPY CONFERENCE

Austin, TX | Tacoma, WA | 2023 - Present

- Directed the **entire editorial pipeline** for three consecutive SciPy Proceedings volumes-issued CFPs, onboarded and managed **150+ reviewers**, and executed double reviews of **30+ submissions** per year, delivering **60+ peer-reviewed papers** and introduced a GitHub Actions build system that cuts production turnaround by **40%**.
- Designed and led **author & reviewer workshops** (rubric-based scoring, reproducibility checks) that drove **95% review completion** and boosted proceedings impact (h-index +15%); partnered with IEEE, Zenodo and CurveNote to secure DOI assignment, open-access release, and indexing for accepted articles.

PROJECTS

- Authored and shipped TikToker, a cross-platform Obsidian community plugin (**1000+ downloads**, active on **iOS, Android, Windows, Linux, macOS**) for saving and reviewing TikTok content as markdown notes; integrated **local Whisper-based ASR inference** (whisper.cpp / faster-whisper) with a yt-dlp + ffmpeg audio processing pipeline for fully on-device transcription, with no external API calls or cloud dependencies.
- Built a session-based recommendation engine using a deep learning-based encoder-decoder architecture incorporating contextual information from users and an attention mechanism, improving displayed recommendations in e-commerce scenarios by 3% over present baselines on the YouChoose dataset (CNARM).
- Developed NLP and ML software to predict outcome of contractual damage lawsuits to benefit academics in debates, parties in drafting contracts, counsels in pre-trial exchanges and judges in rulings.

CLIENT IMPACT AND CONSULTING EXPERIENCE

- Managed 15+ client engagements across the Americas, Middle East, and Asia, delivering AI solutions that generated \$20M+ in measurable business value
- Led technical presentations for C-level executives, translating complex ML concepts into strategic business insights and investment decisions

- Collaborated with multidisciplinary teams of domain experts, engineers, and business analysts across oil & gas, pharma, insurance, retail, and fintech
- Adapted AI solutions to regulated, data-sovereign, and resource-constrained client environments across international markets

PATENTS

US 12,291,957 (Granted May 2025) Field Pump Equipment System (ML-based failure prediction and run-life estimation for ESPs)
US 2025/0270991 (Pending) Field Equipment System (dynacard-based operational condition detection and pump control)
US 2025/0104436 (Pending) Wellsite Operations Machine Vision Framework (line-of-fire and human safety risk detection)
US 2025/0067164 / WO 2023/136856 (Pending) Field Pump Equipment System (real-time pump performance issue detection)
US 2024/0401586 (Pending) Artificial Intelligence-Driven Classification Workflow for Diagnosis of Sucker Rod Pump Operating Conditions
US 2024/0018863 (Pending) Field Equipment System (ML model operationalization and continuous updates on IIoT devices)

PUBLICATIONS

SciPy Proceedings 2025 An Exemplar for Publishing Computational Open Science
2024 ADIPEC Enhancing Edge-Based SRP Production Optimization Algorithm with Fast Loop Mitigation
2024 SPE CETC Real-Time Well Constraint Detection Using an Intelligent Surveillance System
2022 ATCE Real-time Smart Alarms Suite Enabled Through Data Analytics and Edge-based Virtual Flowmeter
2021 SPE Symposium Electrical Submersible Pump Prognostics and Health Monitoring using Machine Learning and NLP

SKILLS

AI/ML Frameworks & Tools	TensorFlow, PyTorch, Keras, Sklearn, OpenCV, Pandas, Spark, MLFlow
Generative AI & LLMs	Dataiku LLM Mesh, Azure AI Foundry, OpenAI API, LangChain, LangGraph, RAG, Fine-tuning, Model Context Protocol (MCP)
Vector Databases	FAISS, ChromaDB, Pinecone, Milvus
Cloud & MLOps	Azure ML, Google Vertex AI, Cognite Data Fusion, Docker, Kubernetes, FastAPI, Git, Azure DevOps, CI/CD
Programming Languages	Python, SQL, R, TypeScript, JavaScript, C++

OPEN SOURCE PROJECTS

OBSIDIAN TIKTOKER  TYPESCRIPT, OBSIDIAN API, WHISPER.CPP, FASTER-WHISPER, PYTHON, FFMPEG

- Authored and shipped a cross-platform Obsidian community plugin (**1000+ downloads**, active on **iOS, Android, Windows, Linux, macOS**) for saving and reviewing TikTok content as markdown notes; integrated **local Whisper-based ASR inference** (whisper.cpp / faster-whisper) with a yt-dlp + ffmpeg audio processing pipeline for fully on-device transcription, with no external API calls or cloud dependencies.

PETROMCP  PYTHON, MODEL CONTEXT PROTOCOL (MCP), LASIO

- Built and open-sourced an **MCP server exposing petroleum data formats** (LAS well logs, with DLIS, SEG-Y, and pump cards on the roadmap) as agent-accessible tools for LLM hosts; **local-first architecture** with strict path allowlisting and zero outbound network calls, shipping well log QC, per-curve statistics, well comparison, and unit conversion tools with a full eval suite and CI; listed on the **MCP.so** community server directory.

SPECWARDEN  PYTHON, CLAUDE CODE HOOKS & SKILLS, PIPX

- Created a **spec-driven development enforcement tool for AI coding agents** like Claude Code and Codex that blocks code edits until a written spec exists and logs every edit with backlinks for auditable agent sessions; hook-based enforcement complementary to GitHub’s spec-kit, with a **published benchmark eval suite** demonstrating a **75%+ reduction in out-of-scope file edits** versus baseline agent runs, cross-platform installers, and coverage reporting.

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE)	Senior Member; Member, IEEE Computer Society
Association for Computing Machinery (ACM)	Member, Artificial Intelligence Special Interest Group (SIGAI)
Society of Petroleum Engineers (SPE)	Member, Data Science and Engineering Analytics and Management

PROFESSIONAL CONTRIBUTIONS AND SERVICE

IEEE Software Engineering Radio (SE Radio), 2025 – present	Host , IEEE Computer Society software engineering podcast
IEEE Scholarship Programs 2026	Judge , IEEE Career Catalyst Scholarship and IEEE-CS Richard E. Merwin Student Scholarship
SciPy Conference 2025	Proceedings Chair , Scientific Computing with Python
SPE ATCE 2024	Chief Judge , International Student Paper Contest
SciPy Conference 2024	Proceedings Chair , Scientific Computing with Python
Optimalex	Senior Technical Advisor , Predictive Analytics
AI in Oil and Gas Conference 2024	Speaker , Energy Conference Network
SciPy 2023	Conference Reviewer , Scientific Computing with Python
Loyola of Los Angeles Int'l and Comparative Law Review (ILR), Vol. 46, No. 1, 2023	Acknowledged Contributor
St. Thomas Journal for Complex Litigation (JCL) 2023	Acknowledged Contributor
Dell Dataiku Data Science Connect 2022	Panelist and Presenter
DataCan 2022	Panelist , Computer Vision through a Magnifying Glass

VOLUNTEERING

SciPy Conference	Organizing Team Member
IEEE Computer Society	AI Standards Committee (C/AISC); Working Groups P3462 (Safety by Design for Generative AI & Child Safety) and P3123 (AI/ML Terminology & Data Formats)
Columbia University CAMP & Odyssey Programs	Mentor, graduate and undergraduate career mentorship (2023 – 2025)
Society of Petroleum Engineers (SPE)	Mentor, Young Professionals Program
Columbia Alumni Association	Organizing Team Member
Columbia University	SLB Campus Recruitment Manager
OIT	Houston Chapter Leadership Team Member
OIT University (OIT-U)	Admissions Team Lead for Diversity, Equity and Inclusion
Houston Humane Society	Pet Adoption Coordinator

AWARDS

Performed by SLB Award 2022	for Technical Excellence and Achievements in Cross-Domain AI/ML Product Development
MLOps World Winner 2021	Dataiku Box of Awesome Challenge
Performed by SLB Award 2021	for Exceptional Development and Management of a High-Priority International Stakeholder Project
Petro.AI Hackathon Silver Award 2020	for Harris County Public Health Hacking for Houston
SLB BeOutstanding Award 2020	for Excellence in Client Engagement and Satisfaction through Building AI and ML Solutions